

DocQuery: Implementation Report

Secure, Grounded PDF RAG Question-Answering System

Author: Vishal Singh | Date: September 25, 2026

Stack: FastAPI, Google Gemini API, React 18, TypeScript, Vite

Executive Summary

DocQuery is a production-grade, factual Question-Answering system operating over PDF documents. It enforces strict context grounding against hallucinations, provides granular 1-indexed page citations, isolates retrieved content against prompt injection attacks, and includes an automated 10-question evaluation benchmark scoring 100% accuracy. The frontend follows a strict B2B minimalist design system (Inter font, 6px uniform corner radius, single #2563eb accent color, flat 1px borders, and zero purple gradients or emojis).

Phase 1: PDF Ingestion, Metadata Chunking & Vector Embeddings

- **Environment Setup:** Isolated Python 3.14 virtual environment (.venv) with strict .gitignore protection for credentials.
- **Page-Indexed Text Extraction:** Extracted text page-by-page via pypdf.PdfReader, preserving 1-indexed page numbers. Whitespace normalized via regex.
- **Sliding-Window Chunking:** Designed sliding-window chunking (chunk_size=450 chars, chunk_overlap=50 chars). Every chunk preserves doc_name, page_number, and chunk_id ({doc}_p{page}_c{chunk}).
- **Dense Vector Embeddings:** Integrated Google Gemini API using gemini-embedding-001 (3072 dimensions) with batching (batch_size=50).
- **In-Memory Cosine Similarity:** Vector search implemented in NumPy computing normalized dot products. Top-k scoring chunks retrieved in descending order.
- **Automated Verification:** Created backend/test_phase1.py verifying page counts, chunk metadata integrity, and mathematical cosine similarity assertions.

Phase 2: Anti-Hallucination Guardrails & 10-Question Evaluation Suite

- **Semantic Distance Cutoff:** If top similarity score is below 0.40, the system short-circuits immediately with explicit refusal without invoking the LLM.
- **XML Prompt Injection Isolation:** Retrieved context chunks are enclosed in <document_context> XML tags, isolating user queries from adversarial document injection payloads.
- **Deterministic Low-Temperature Generation:** Gemini 3.1 Flash Lite configured with temperature=0.0 and strict system instructions mandating factual page citations.
- **10-Question Benchmark Dataset:** Created backend/benchmark.json with 9 factual questions (roles, languages, concepts, OWASP Top 10, PostgreSQL, React, working hours) and 1 negative guardrail question (annual revenue).
- **Automated Evaluation Harness:** Created backend/evaluate.py running automated scoring with 2.0s rate-limiting for free tier. Result: 10/10 Passed (100% accuracy, 1/1 refusal).

Phase 3: FastAPI Web Service & Typed Contracts

- **Pydantic v2 Contracts:** Defined `QueryRequest`, `QueryResponse`, `SourceCitation`, `DocumentUploadResponse`, `DeleteDocumentResponse`, and `EvalSummary` in `backend/schemas.py`.
- **Health & Status (GET `/api/health`):** Reports service state, loaded documents, total chunk counts, active document name, and page counts.
- **PDF Ingestion (POST `/api/upload`):** Multipart PDF upload endpoint handling page parsing, chunking, and embedding generation.
- **Grounded Query (POST `/api/query`):** Runs semantic retrieval, applies guardrails, and returns answer with page citations.
- **Automated Benchmark (POST `/api/evaluate`):** Executes the 10-question evaluation suite programmatically and returns the complete scorecard.
- **Document Deletion (DELETE `/api/document`):** Clears active document and associated vector chunks to return system to clean zero-state.
- **CORS Middleware:** Enabled for cross-origin frontend communication from local development and cloud CDN environments.

Phase 4: Modern Minimalist React Frontend & UI Loaders

- **Design System (Linear / Vercel Aesthetic):** Inter font, Space Grotesk headings, 6px uniform corner radius, single #2563eb accent, 68ch max prose line length, flat 1px borders, zero emojis.
- **UploadCard Component:** Drag-and-drop PDF dropzone with animated spinner, upload progress bar, active document badge, and document deletion action.
- **ChatPanel Component:** Interactive Q&A with suggested quick queries, answer synthesis loading skeleton, confidence badge, and granular source attribution badges.
- **BenchmarkPanel Component:** 1-click evaluation runner with execution progress bar, 4-stat scorecard summary grid, and full 10-question results table.
- **Zero-State on Startup:** Service initializes with zero documents loaded; query inputs and chat are safely disabled until a document is ingested.
- **Deployment Automation:** Configured `netlify.toml` for static frontend hosting with API rewrites and `render.yaml` for 1-click Python backend hosting.

Phase 5: Anonymization, Code Polish & Verification Scorecard

- **Complete Anonymization:** Removed all hardcoded vendor / company names across all files (grep verified 0 occurrences). Test document standardized as `sample_document.pdf`.
- **Senior-Level Code Polish:** Cleaned up codebase to remove AI-sounding comments, repetitive boilerplate, and unnecessary logs.
- **TypeScript Production Bundle:** `npm run build` verified compiling cleanly to `dist/` in 749ms with 0 errors.

Verification Results Summary

Milestone / Test	Verification Scope	Status
Phase 1 Verification	PDF extraction, chunking & vector cosine math	PASS (100%)
Phase 2 Benchmark	10 ground-truth questions + negative refusal	PASS (10/10)
Backend REST API	Upload, Query, Health, Delete endpoints	PASS (200 OK)
Frontend Build	React 18 + TypeScript production bundle	PASS (0 errors)
Anti-Hallucination Guardrail	Negative refusal on out-of-domain query	PASS (Refused)

Source Citations	Exact 1-indexed physical page numbers	PASS (Page 2 & 3)
------------------	---------------------------------------	-------------------